OSSD - Calculus and Vector

MCV4U

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Unit 1: Rate of Change

In this unit, we study rates of change, continuity, and limits, followed by an introduction to derivatives.

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1 Slope

2 Derivatives

2.1 Introduction to Derivatives

Definition 2.1 (Derivative)**.** The derivative measures the instantaneous rate of change of a function with respect to its input. Geometrically, it is the slope of the tangent line to the graph at a point.

Common notations are

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

$$m_T = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h},$$

$$\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

Useful notation:

- Average rate of change over an interval: $\dfrac{\Delta y}{\Delta x}$.
- Instantaneous rate of change at a point: $\dfrac{dy}{dx}$.

For a function $f(x)$ at $x = a$,

$$\left.\frac{dy}{dx}\right|_{x=a} = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

and equivalently

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}.$$

Another equivalent form is

$$\lim_{x\to a} \frac{f(x) - f(a)}{x - a}.$$

Example 2.2. Given $f(x) = \dfrac{1}{x}$, find the slope of the tangent at $x = 2$.

$$\begin{aligned} m &= \lim_{x\to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} \\ &= \lim_{x\to 2} \frac{2 - x}{2x(x-2)} \\ &= \lim_{x\to 2} \left(-\frac{1}{2x}\right) \\ &= -\frac{1}{4}. \end{aligned}$$

The derivative may fail to exist at $x = a$ when:

- the function is discontinuous at $x = a$;
- the function has a corner at $x = a$;
- the function has a cusp at $x = a$;
- the tangent is vertical at $x = a$.

3 Differentiation Rules

3.1 Derivatives of Polynomial Functions

3.1.1 Power Functions

A power function has the form

$$x^a, \quad x \in \mathbb{R},\ a \in \mathbb{R},$$

where x is the base and a is the exponent.

3.1.2 Power Rule

If

$$f(x) = x^n,$$

then

$$f'(x) = nx^{n-1}.$$

In words: move the exponent to the front as a coefficient, then subtract 1 from the exponent.

Remark. Do not apply the power rule to exponential functions such as a^x (for example, 2^x).

3.1.3 Basic Derivative Rules

$$\begin{aligned}
\text{If } f(x) = c, &\quad f'(x) = 0,\\
\text{If } f(x) = x^n, &\quad f'(x) = nx^{n-1},\\
\text{If } f(x) = c\,g(x), &\quad f'(x) = c\,g'(x),\\
\text{If } h(x) = f(x) + g(x), &\quad h'(x) = f'(x) + g'(x),\\
\text{If } h(x) = f(x) - g(x), &\quad h'(x) = f'(x) - g'(x).
\end{aligned}$$

3.2 The Product Rule

Not every limit rule turns directly into a derivative rule. A common mistake is

$$\text{if } g(x) = f(x)h(x), \text{ then } g'(x) = f'(x)h'(x),$$

which is false in general.

Proposition 3.1 (Product Rule)**.** *If $g(x) = f(x)h(x)$, then*

$$g'(x) = f'(x)h(x) + f(x)h'(x).$$

Proof. Using first principles,

$$\begin{aligned}
g'(x) &= \lim_{\Delta x \to 0} \frac{f(x+\Delta x)h(x+\Delta x) - f(x)h(x)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \frac{f(x+\Delta x)h(x+\Delta x) - f(x)h(x+\Delta x)}{\Delta x} \\
&\quad + \lim_{\Delta x \to 0} \frac{f(x)h(x+\Delta x) - f(x)h(x)}{\Delta x} \\
&= \lim_{\Delta x \to 0} \left(h(x+\Delta x) \frac{f(x+\Delta x) - f(x)}{\Delta x} \right) \\
&\quad + \lim_{\Delta x \to 0} \left(f(x) \frac{h(x+\Delta x) - h(x)}{\Delta x} \right) \\
&= h(x)f'(x) + f(x)h'(x).
\end{aligned}$$

□

4 Reference Sheet

4.1 Assumed Knowledge

4.1.1 Log Laws

$$\log_b A + \log_b B = \log_b(AB),$$
$$\log_b\left(\frac{M}{N}\right) = \log_b M - \log_b N,$$
$$\log_b(M^k) = k\log_b M,$$
$$\log_b(b^k) = k,$$
$$b^{\log_b k} = k.$$

4.1.2 Trig Laws

$$\tan\theta = \frac{\sin\theta}{\cos\theta},$$
$$\cot\theta = \frac{\cos\theta}{\sin\theta},$$
$$\csc\theta = \frac{1}{\sin\theta},$$
$$\sec\theta = \frac{1}{\cos\theta},$$
$$\sin^2\theta + \cos^2\theta = 1,$$
$$1 + \cot^2\theta = \csc^2\theta,$$
$$\tan^2\theta + 1 = \sec^2\theta,$$
$$\sin(A \pm B) = \sin A\cos B \pm \cos A\sin B,$$
$$\cos(A \pm B) = \cos A\cos B \mp \sin A\sin B,$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}.$$

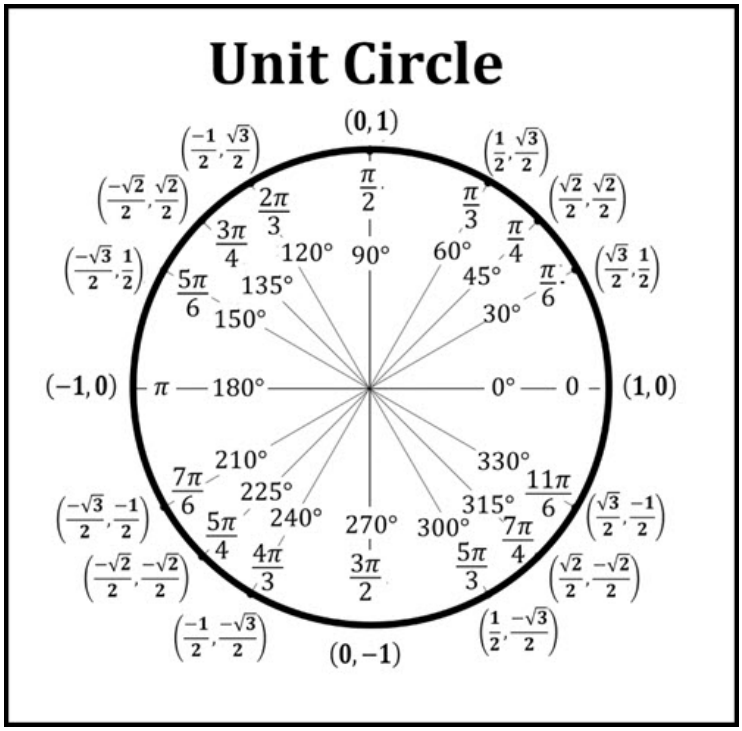
Sine law:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

Cosine law:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

Unit circle:


Unit Circle
(0, 1)
(1, 0)
(0, −1)
(−1, 0)
π/2
90°
π/3
60°
π/4
45°
π/6
30°
0°
0
2π/3
120°
3π/4
135°
5π/6
150°
π
180°
7π/6
210°
5π/4
225°
4π/3
240°
3π/2
270°
5π/3
300°
7π/4
315°
11π/6
330°
(1/2, √3/2)
(√2/2, √2/2)
(√3/2, 1/2)
(−1/2, √3/2)
(−√2/2, √2/2)
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